Model input mapping
Supporting Document: `IndigoCarbon_US-1_2023_0030`

Contents

1 Management-Based Inputs **2**
- Special Rule for Planting: Interseeding . . . 4
- Special Rule for Planting: Multiple Plantings on Same Day . . . 5
- Special Rule for Planting: Cover Crop Blends . . . 5
- Harvest . . . 6
  - Parameter Derivations . . . 6

N Fertilizer Application **7**
- Parameter Derivations . . . 7
- Special Rule for N Fertilizer Application: Multiple Applications on Same Day . . . 8
- Organic Amendment Application . . . 8
  - Parameter Derivations . . . 8
  - Special Rule for Organic Amendment Application: Multiple Applications on Same Day . . . 9
  - . . . 9
- Tillage . . . 9
  - Parameter Derivations . . . 9
  - Derivation of the STIR Score . . . 11
  - Derivation of the effective depth of disturbance . . . 12
  - Special Rule for Tillage: Tillage at Planting for No-Till . . . 12
  - Special Rule for Tillage: No Tillage Event Within 30 Days of Planting . . . 12
  - Special Rule for Tillage: Tillage and Harvest Events on the Same Day . . . 13
  - Special Rule for Tillage: Herbicide to Kill a Crop . . . 13
  - Special Rule for Tillage: Root Crops . . . 13
  - Special Rule for Tillage: Multiple Tillage Events on Same Day . . . 13
- Irrigation . . . 15
  - Parameter Derivations . . . 15
- Burning . . . 15
  - Parameter Derivations . . . 15
- Model Spin-up . . . 15
  - Model Runs to Establish Conservative Defaults . . . 16

Non-Management-Based Inputs **21**
- Weather Inputs . . . 21
- Soil Inputs (Data Conversions) . . . 22
  - . . . 24
  - Bulk Density . . . 24
  - . . . 25
  - References . . . 25

2 Summary of Changes **26**

This document describes how each input to the biogeochemical model DayCent-CR Version 1.1.0 was derived using Grower Survey answers, soil measurement data, and other sources (see MP Table 6.2 for a list of all model inputs,

and `IndigoCarbon_US-1_2023_0029` for the list of Grower Survey questions).

1 Management-Based Inputs

The input parameters related to planting include CropNumber, CropName, ContinueFromPreviousYear, and PlantingDate. CropNumber is assigned an integer starting with 1 by an increment of 1 based on the number of crops being planted in a given calendar year. For example, if two cash crops are followed by a cover crop, assign CropNumber of 1 and 2 for the first and second cash crop and assign 3 for the cover crop.

CropName is mapped from the Grower Survey questions "What crop did you plant in this field?" and "What cover crop species did you plant in this field?" Crops were mapped to exact matches in DayCent-CR when such matches existed, and substitutions were used in all other cases. All substitutions were made using crops within the same Crop Functional Group, as permitted by the SEP Model Requirements and Guidance v1.1a. Mappings between the Grower Survey responses and DayCent-CR options are provided in Table 1 below, along with the Crop Functional Group (CFG) as defined by the SEP Model Requirements and Guidance v1.0a.

PlantingDate was obtained directly from the Grower Survey question "When was this crop planted?" ContinueFromPreviousYear was set to "Yes" only when intrinsically perennial crops were planted in a previous year and continued into the current year. Note that "perennial" crops as defined by SEP Model Requirements and Guidance v1.1a relates to Crop Functional Groups used in model validation, and is based on the USDA Natural Resources Conservation Service (NRCS) definitions of annual/perennial designation for individual crops (*2014 Farm Bill - Conservation Compliance Crop List | NRCS*, 2014). Per the NRCS, alfalfa annual/perennial status is locally determined.

Table 1: Mappings for CropName. Substitutions in the middle column are indicated with parentheses.

Grower response to "What crop did you plant in this field?"	**Value assigned to CropName in DayCent-CR**	**Crop Functional Group (CFG)**	**Cover crop type**
Alfalfa	Alfalfa	N-fixing, annual, C3, herbaceous, non-flooded, aka "SOY"	n/a
Barley	Barley	Non-N-fixing, annual, C3, herbaceous, non-flooded, aka "WHEAT"	n/a
Beans/Peas	Dry Field Beans	SOY	n/a
Canola	Sunflower	WHEAT	n/a
Corn	Corn	Non-N-fixing, annual, C4, herbaceous, non-flooded, aka "CORN"	n/a
Cotton	Cotton	COTTON	n/a
Flax	Sunflower	WHEAT	n/a
Oats	Oats	WHEAT	n/a
Peanuts	Annual Peanut	SOY	n/a
Rye	Rye	WHEAT	n/a
Sorghum	Sorghum	CORN	n/a
Soybeans	Soybeans	SOY	n/a
Sugar Beets	Sugar Beets	WHEAT	n/a
Sunflowers	Sunflower	WHEAT	n/a
Wheat	Winter Wheat or Spring Wheat*	WHEAT	n/a
Alfalfa	Alfalfa	SOY	legume
Alfalfa, annual	Alfalfa	SOY	legume
Alfalfa, grazing	Alfalfa	SOY	legume
Alkali Grass	(Annual Rye)	WHEAT	legume
Alsike Clover	Clover	SOY	legume
Australian winter peas	Austrian Winter Pea	SOY	legume

Table 1: Mappings for CropName. Substitutions in the middle column are indicated with parentheses.

Grower response to "What crop did you plant in this field?"	**Value assigned to CropName in DayCent-CR**	**Crop Functional Group (CFG)**	**Cover crop type**
Balansa Clover	Clover	SOY	legume
Barley	Barley	WHEAT	grass
Bentgrass	(Annual Rye)	WHEAT	grass
Berseem Clover	Clover	SOY	grass
Birdsfoot trefoil	Clover	SOY	legume
Bluegrass	Kentucky Bluegrass	WHEAT	grass
Bromegrass	Brome grass	WHEAT	grass
Buckwheat	(Annual Rye)	WHEAT	broadleaf
Canadian forage pearl millet	Millet	CORN	grass
Cereal rye	Cereal Rye	WHEAT	grass
Chickpea	Dry Field Pea	SOY	legume
Chicory	(Annual Rye)	WHEAT	broadleaf
Cicer Milkvetch	Vetch	SOY	legume
Clovers	Clover	SOY	legume
Cover crop blend**	Three species mix	SOY	legume, grass
Cover crop blend**	Two species mix	SOY, WHEAT	legume, grass
Crimson clover	Clover	SOY	legume
Crown vetch	Vetch	SOY	legume
Cowpea	Dry Field Pea	SOY	legume
Daikon radish	Forage Radish	WHEAT	broadleaf
Faba beans	Dry Field Beans	SOY	legume
Fenu greek	(Annual Rye)	WHEAT	legume
Fescues	Idaho Fescue	WHEAT	grass
Field peas	Dry Field Peas	SOY	legume
Flax	(Sunflower)	WHEAT	broadleaf
Fodder rape	Forage Radish	WHEAT	brassica
Forage chicory	(Annual Rye)	WHEAT	broadleaf
Forage peas	Dry Field Peas	SOY	legume
Forage radish	Forage Radish	WHEAT	brassica
Forage rape	Forage Radish	WHEAT	brassica
Galega	(Annual Rye - Legume)	SOY	legume
German millet	Millet	CORN	grass
Grazing turnip	Forage Radish	WHEAT	brassica
Hairy vetch	Vetch	SOY	legume
Hemp, sun	(Annual Rye - Legume)	SOY	legume
Japanese millet	Millet	CORN	grass
Kale	(Annual Rye)	WHEAT	brassica
Ladino clover	Clover	SOY	legume
Lentil	(Vetch)	SOY	legume
Milk vetch	Vetch	SOY	legume
Millet	Millet	CORN	grass
Mixtures, various	Grass	WHEAT	n/a
Mustard	Forage Radish	WHEAT	brassica
Oats	Oats	WHEAT	grass
Oilseed radish	Oilseed Radish	WHEAT	brassica
Orchard grass	Orchard grass	WHEAT	grass
Pearl millet	Millet	CORN	grass
Peas	Dry Field Pea	SOY	legume

Table 1: Mappings for CropName. Substitutions in the middle column are indicated with parentheses.

Grower response to "What crop did you plant in this field?"	**Value assigned to CropName in DayCent-CR**	**Crop Functional Group (CFG)**	**Cover crop type**
Persian clover	Clover	SOY	legume
Phacelia	(Annual Rye)	WHEAT	broadleaf
Radish	Forage Radish	WHEAT	brassica
Rape	Forage Radish	WHEAT	brassica
Red clover	Clover	SOY	legume
Reed Canary Grass	(Annual Rye)	WHEAT	grass
Rye	Rye	WHEAT	grass
Rye triticale	(Annual Rye)	WHEAT	grass
Ryegrass, annual	Annual ryegrass	WHEAT	grass
Ryegrass, perennial	(Annual ryegrass)	WHEAT	grass
Sorghum	Sorghum	CORN	grass
Spring oats	Oats	WHEAT	grass
Spring peas	Dry Field Pea	SOY	legume
Strawberry Clover	Clover	SOY	legume
Sudangrass	Sorghum	CORN	grass
Sugar beet	Sugar Beets	WHEAT	brassica
Sunflower	Sunflower	WHEAT	broadleaf
Sweet clover	Clover	SOY	legume
Teff grass	(Millet)	CORN	grass
Tillage Turnip	Forage Radish	WHEAT	brassica
Timothy	timothy	WHEAT	grass
Toro radish	Forage Radish	WHEAT	brassica
Trefoil	Clover	SOY	legume
Triticale	Winter Wheat	WHEAT	grass
Turnip	Forage Radish	WHEAT	brassica
Vetch	Vetch	SOY	legume
Vetch fescue	Vetch	SOY	legume
Wheat, winter or spring	Winter Wheat	WHEAT	grass
White Clover	Clover	SOY	legume
White lupins	Clover	SOY	legume
White mustard	Forage Radish	WHEAT	brassica
White proso millet	Millet	CORN	grass
Winter barley	Barley	WHEAT	grass
Winter peas	Dry Field Pea	SOY	legume
Winter rye	Rye	WHEAT	grass
Winter triticale	Winter Wheat	WHEAT	grass
Yellow-white mustard	Forage Radish	WHEAT	brassica

*CropName is assigned to "Winter Wheat" if the Field Manger indicates "Wheat" was grown, and the planting and harvest dates do not occur in the same calendar year. Otherwise it is assigned to "Spring Wheat"

**See Special Rule for Planting: Cover Crop Blends below

Special Rule for Planting: Interseeding

Interseeding of crops is not permitted in DayCent-CR and will result in a simulation failure if crops overlap, i.e. if a new crop is planted before the previous crop has been harvested or terminated. It is therefore necessary to manually separate overlapping crop seasons before crediting simulations are performed. There are five scenarios to consider for interseeded crops:

1. Cover crop planted into a cash crop

- If the cover crop is planted after the maturity date of the cash crop, the cover crop planting date is retained and the cash crop is harvested.
  - If the cash crop harvest is for grain (i.e. "Grain" = "Yes"), then the cash crop harvest is performed one day prior to the cover crop planting date.
  - If the cash crop is not harvested for grain, i.e. "Grain" = "No", then the cash crop is harvested two days prior to the cover crop planting date to allow for the killing of the cash crop using a tillage with herbicide event (see *Special Rule for Tillage: Killing with Herbicide*).
- If the cover crop is planted before the cash crop maturity date, the cash crop is harvested the day after the cash crop maturity date.
  - If the cash crop is harvested for grain, the cover crop is planted one day after the cash crop maturity date.
  - If the cash crop is not harvested for grain, the cover crop is planted two days after the cash crop harvest to allow for the killing of the cash crop with a tillage with herbicide event (see *Special Rule for Tillage: Killing with Herbicide*).

2. Cash crop planted into a cover crop (i.e., "planting green")

- Retain the cash crop planting date.
  - If the cover crop is harvested for grain, terminate the cover crop one day prior to the cash crop planting date.
  - If the cover crop is not harvested for grain, terminate the cover crop two days prior to cash crop planting date to allow for the killing of the cover crop with a tillage with herbicide event (see *Special Rule for Tillage: Killing with Herbicide*).

3. Cash or cover crop season is enveloped by the other

- Grow only the cash crop; retain both the planting and harvest dates.

4. Cover crop planted into a cover crop

- If both cover crops have the same legume or non-legume status, grow a single cover crop using the planting date of the first cover crop and termination date of the second. Set the species to whichever cover has the longer season.
- If the legume status changes between crops, terminate the first cover crop two days before planting of the second to allow for the killing of the first cover crop with a tillage with herbicide event (see *Special Rule for Tillage: Killing with Herbicide*) the day after termination. Preserve the planting date of the second cover crop.

5. One cover crop season is enveloped by another

- Grow only the cover crop with the longer season.
  - If either cover crop is a legume that is grown for at least three weeks, set the species to that legume species.
  - If both species are legumes, choose the legume species with the longer season.

Special Rule for Planting: Multiple Plantings on Same Day

DayCent-CR will not simulate more than one planting event on the same day. It will only use the last such event provided for a given day. Since interseeding is not allowed, two planting events on the same day should never occur.

Special Rule for Planting: Cover Crop Blends

If the number of cover crops in the mix is 1, then map the cover crop directly to the associated DayCent-CR CropName.

Otherwise, if the number of cover crops in the mix is greater than 1:

- For cover crops in the mix, add up the % fraction in each of the cover crop species type classes (i.e., GRASS, LEGUME, BRASSICA, BROADLEAF, NOT_APPLICABLE)

- If the number of cover crop species type classes in the mix is equal to 1:
  - If one of the crops in the mix is designated as dominant, use that as the crop type to represent the mix.
  - Otherwise, if one of the crops in the mix has a higher % of the fraction in the mix then use that as the crop type to represent the mix
  - Otherwise, if the % of the fraction in the mix is the same for all crops in the mix, sort the crops alphabetically by name and use the first crop in the list as the crop type to represent the mix
- If the number of cover crop classes in the mix is greater than 1:
  - When the fraction of legume is greater than zero, model the following cover crop blends in DayCent_CR:
    * Three species mix for 3 or more cover crop classes
    * Two species mix for 2 cover crop classes
  - When the fraction of legume is zero and grass and [BRASSICA, BROADLEAF, or NOT_APPLICABLE] are present in the mix, model as Cereal Rye

Harvest

Parameter Derivations

The input parameters related to harvest include HarvestDate, Grain, and StrawStoverHayRemoval. HarvestDate is obtained directly from the Grower Survey question "When was this crop harvested?". Both Grain and StrawStoverHayRemoval are assigned based on responses to the Grower Survey question "How was the crop harvested?", as indicated in Table 2. Additionally, during root crop (i.e., Peanuts and Sugar Beets) harvesting, 90% of below ground live biomass is removed and a separate Reduced Tillage event is added to ensure an appropriate amount of soil disturbance occurs due to root harvesting.

Table 2: Mappings of Grain and StrawStoverHayRemoval.

Grower response to "How was the crop harvested?"	**Value for DayCent-CR "Grain" input parameter**	**Value of DayCent-CR "StrawStoverHayRemoval" input parameter (%)**
Grain only	Yes	0
Grain with subsequent straw, stover, or residue removal	Yes	50
Bailed straw	No	30
Silage	No	90
Haylage	No	80
Hay	No	80
Root	Yes	0

While a "Grain only" harvest is self-explanatory in its assignment of zero removal of straw, stover, and hay, a harvest of "Grain with subsequent straw, stover, or residue removal" required a review of literature, extension resources, and other sources to obtain an average value to be used in modeling simulations. Ultimately, a value of 50% was selected as an average across these sources. Sources included the following:

- A publication by the Iowa Corn Growers Association stating that "Typically, in high-yielding fields managed with minimum or no-till practices, about 45 percent, or two tons per acre, of the stover can be sustainably removed following each corn crop." (Sustainable Corn Stover Harvest, 2013).
- A global meta-analysis of bioenergy studies looking at the effect of corn stover removal on soil carbon cited an average stover removal of 71% in longitudinal studies, and 74% in paired studies (Xu et al. 2019).
- A University of Wisconsin–Madison extension publication states that "Even with the best methods you will only be able to harvest 50-80% of stover" (Mayer 2012).

- A USDA–ARS publication indicated that for corn yields between 150–200 bu/ac, 23-43% of stover can be safely removed (Karlen and Huggins, 2014).
- A US–DOE report stated that for crops harvested for feedstock, such as corn, "Collection of residues from these crops involves multiple passes of equipment over fields and results in no more than 40 percent removal of stover or straw on average. This low recovery amount is due to a combination of collection equipment limitations, contour ridge farming, economics, and conservation requirements. It is possible under some conditions to remove as much as 60-70 percent of corn stover with currently available equipment." (Perlack et al. 2005)

For "Bailed straw", two sources were used to derive an average value of 30% to be used in modeling simulations:

- A 2009 publication in Agronomy Journal focused on the effects of straw removal via baling on long term soil quality and wheat production cited that "The proportion of total aboveground residues other than grain removed with baling ranged from 22 to 35% or 26 to 40% depending on the method of calculation based on the first study." (Lafond et al. 2009)
- A US–DOE reports that "Collection of residues from these crops involves multiple passes of equipment over fields and results in no more than 40 percent removal of stover or straw on average. This low recovery amount is due to a combination of collection equipment limitations, contour ridge farming, economics, and conservation requirements. It is possible under some conditions to remove as much as 60-70 percent of corn stover with currently available equipment." (Perlack et al. 2005)

For "Silage", three sources were used to derive an average of 90% to be used in modeling simulations:

- Bolinder et al. (1999) who cited that "In the case of silage corn, only 5-10% is left as stubble."
- A cut height of ~7" for corn silage reported by Penn State Extension (Roth 2016), combined with an average height of a corn plant at harvest of ~96" given by USDA-NASS Quick Stats (IndigoCarbon_US-1-2023_0069). The approximate percentage change would thus be (96 – 7) / 96 * 100 = 93%
- Subject matter expert approval from Soil Metrics staff, who cite 90% as the value used in the US EPA National Greenhouse Inventory (USEPA 2021) model simulations using DayCent for silage harvests.

For both "Hay" and "Haylage", we followed guidance from subject matter experts at Soil Metrics who cited 80% as the removal rates used for both "Hay" and "Haylage" harvests in the US EPA National Greenhouse Gas Inventory (USEPA 2021).

N Fertilizer Application

Parameter Derivations

The input parameters related to N fertilizer application include NApplicationDate, NApplicationType, and NApplicationAmount. The DayCent-CR API requires non-null values also be provided for two additional parameters, NApplicationMethod and EEP, however neither of these are passed on to DayCent-CR as the model currently lacks the functionality to represent them in simulations or the functionality is not available in the API.

NApplicationDate is obtained directly from the Grower Survey question "On what date did you apply this fertilizer?". NApplicationType refers to the form of the nitrogen applied. The DayCent-CR API provides 21 forms to choose from, however each form is ultimately converted into its relative fractions of ammonium (NH_4) and nitrate (NO_3), the only nitrogen species required by DayCent-CR. We have chosen to directly provide the API with the amounts of pure NH_4 and NO_3, using the API options "Calcium Nitrate" for NO_3, and "Urea" for NH_4. This means that each form of N fertilizer applied is converted into the corresponding proportions of each species. The specific fractions are derived from responses to the Grower Survey questions in Table 3 (IndigoCarbon_US-1_2023_0029). Before indicating their specific product, the grower must distinguish their fertilizer as being either a (common) bulk fertilizer, a specialty product, or a custom blend.

Table 3: Grower survey questions and options used to derive NApplicationType input parameter to DayCent-CR.

Parameter	Grower Survey Question	Grower Survey Options
NApplicationType	What bulk fertilizer product was applied?	SME-established list of common fertilizer products (Table 3.2)
	What specialty fertilizer product was applied?	Crop Data Management Systems (CDMS) database of products (IndigoCarbon_US-1_2023_0069)
	What custom blend fertilizer product was applied?	User may specify their own NPKS blends, including micronutrients
	If custom fertilizer was applied, please specify the form of nitrogen.	Ammonium Ammonium nitrate Anhydrous ammonia Nitrate Urea

Table 4: Relative fractions of NO_3 and NH_4 for some common fertilizer products.

Nitrogen Fertilizer Product	Fraction NH_4	Fraction NO_3
Ammonium Nitrate (34-0-0)	0.5	0.5
Ammonium Polyphosphate Solution (10-34-00)	1.0	0.0
Anhydrous Ammonia (82-0-0)	1.0	0.0
Calcium Nitrate	0.0	1.0
Diammonium Phosphate (18-46-0)	1.0	0.0
Potassium Nitrate	0.0	1.0
Urea (46-0-0)	1.0	0.0
Urea Ammonium Nitrate (30-0-0)	0.75	0.25

NApplicationAmount is provided to the DayCent-CR API in units of lbs N ac^{-1}, and is obtained from the Grower Survey question "What was the fertilizer application rate? (per unit & area)". Weight of fertilizer is converted to weight of N based on the percentage N in the fertilizer. For example, urea (46-00-00) contains 46% N by weight, so 100 lbs of urea ac^{-1} would be converted to 46 lbs of N ac^{-1}.

Special Rule for N Fertilizer Application: Multiple Applications on Same Day

DayCent-CR will not simulate more than one N fertilizer application event on the same day. It will only use the last such event provided for a given day. To accommodate this limitation and ensure all N fertilizer is applied as close to the application date as possible, applications of NH_4 (using NApplicationType = "Urea") and NO_3 (using NApplicationType = "Calcium Nitrate") are made on separate consecutive days, beginning on the date of application (NApplicationDate).

Organic Amendment Application

Parameter Derivations

The input parameters related to organic matter application include OMADAmount, OMADApplicationDate, and OMADCNRatio. The DayCent-CR API requires non-null values also be provided for two additional parameters, OMADType and OMADPercentN (% total or sum of inorganic and organic N), however neither of these are used by DayCent-CR.

OMADAmount, in units of tons dry matter per acre, is calculated using the application rate and type of organic amendments obtained from the Grower Survey questions "What was the organic amendment application rate?" and "What type of organic amendment was applied?" and the corresponding % moisture content in Table 5.

OMADApplicationDate is obtained directly from the Grower Survey question "On what date did you apply this organic amendment?". Table 5 contains values of OMADCNRatio for each of the organic amendment type options listed in

the grower survey. The organic amendment types listed in the Carbon UI dropdown (IndigoCarbon-1_2023_0029) are taken from COMET-Planner (http://comet-planner.com/, Swan et al. 2020), as are the values of C:N ratio.

Table 5: Characteristics of each organic amendment type derived from COMET-Planner.

Response to Grower Survey Question "What type of organic amendment was applied?"	**C:N Ratio (OMADCNRatio)**	**% Moisture**	**% Total N (OMADPercentN)**
Beef feedlot manure - Slurry	15.6	91	0.32
Beef feedlot manure - Solid	11.1	85	0.45
Bloodmeal - Solid	4	15	10.3
Cattle Manure - Solid	11.7	45	1.2
Chicken broiler manure - Solid	7.8	74	2.3
Chicken layer manure - Slurry	4.3	87	0.23
Chicken layer manure - Solid	7.1	75	1.7
Dairy manure - Slurry	7.7	92	0.13
Dairy manure - Solid	12	82	0.5
Feathermeal - Solid	4.8	15	11.6
Fishmeal - Solid	2.25	50	7.85
Swine manure - Slurry	5.9	96	0.17
Swine manure - Solid	6.2	82	0.65
Compost or composted manure	18	50	1

Special Rule for Organic Amendment Application: Multiple Applications on Same Day

DayCent-CR will not simulate more than one organic amendment application event on the same day. It will only use the last such event provided for a given day. To accommodate this limitation, multiple organic amendments occurring on the same day are combined into a single event using the following logic:

- OMADApplicationDate is the application date for all events;
- OMADAmount is the sum of all OMADAmounts in units of tons dry matter per acre.
- OMADCNRatio is the mass-weighted average of C:N ratio for the individual organic amendments applied:
  - $\text{OMADCNRatio} = \text{Weight}_1(\text{OMADCNRatio}_1) + \text{Weight}_2\,(\text{OMADCNRatio}_2) + \ldots\ \text{Weight}_\text{N}(\text{OMADCNRatio}_\text{N})$, where N is the number of applications made on the OMADApplicationDate.
  - $\text{Weight}_\text{N} = \text{OMADAmount}_\text{N}$ / (sum of all OMADAmounts)

Tillage

Parameter Derivations

The input parameters for tillage events include TillageDate and TillageType. TillageDate is obtained directly from the Grower Survey question "On what date did tillage occur?". The derivation of TillageType is based on responses to the Grower Survey question "What tillage method / equipment was used?". The responses to this question are then mapped to the options for TillageType in DayCent-CR, which include "Intensive Tillage", "Reduced Tillage", "Mulch Tillage", "Ridge Tillage", "Strip Tillage", "No Tillage", "Growing Season Cultivation", "Mow", "Crimp", and "Broad-spectrum herbicide" (see Table 6). The DayCent-CR options are then mapped by the DayCent-CR API to specific cultivation events ("CULT" events) in DayCent-CR, labeled A–K. CULT events span a spectrum of soil disturbance captured numerically in DayCent-CR via a cultivation factor that ranges from 0–1. CULT A has the least amount of disturbance and therefore has the lowest cultivation factor. At the other end of the spectrum is CULT K which has the most amount of disturbance and thus is the event with the highest cultivation factor, e.g. a moldboard plow. Note that a continuous range of cultivation factors is not available via the API to support simulations in DayCent-CR; only a subset of the discrete options A–K are available. These include A, B, C, D, F, H, and K. Lastly, CULT events "Broad-spectrum herbicide" do not disturb the soil or have any effect on soil decomposition, but do

move carbon between pools. For example, CULT "Broad-spectrum herbicide" moves all standing live biomass to the standing dead pool in DayCent-CR. Similarly, CULT "Mow" will cut the plant and move 75% of above-ground biomass to surface litter. While CULT "Mow" and "Crimp" are both available within the DayCent-CR API, neither event is currently used.

The hierarchy for additional tillage practices is based on the NRCS Soil Tillage Impact Rating (STIR) score for a given implement or method in the UI (described in `IndigoCarbon_US-1_2023_0024a`). A high impact category includes any STIR score > 45, a medium impact category includes any STIR score between 25 and 45, and low includes any STIR score < 25. A switch from a higher impact category to a lower impact category is considered additional. The derivation of the STIR score for each implement and method is provided in the next section.

To conform with the requirements of DayCent-CR, it was necessary to formulate mappings between both the impact categories and the methods / implements to the CULT events available for use in DayCent-CR. First, the CULT options (A–K) were compared with the STIR score-based impact categories, and CULT events placed within each category based on Indigo SME guidance. This resulted in one or more CULT events being assigned to each category (table 6).

Table 6: TillageType Options in DayCent-CR API with corresponding CULT events and cultivation factors in DayCent-CR. TillageType options that are not available or not used via the DayCent-CR API are labeled as such in the first column.

DayCent-CR TillageType Option in API	**DayCent-CR CULT Event**	**DayCent-CR Cultivation Factor Range**	**Impact Category**
(Not used)	A	0.001–0.02	Very low impact
No tillage - Planter	B	0.02–0.06	Low impact
Strip Tillage and Growing Season Cultivation	C	0.06–0.12	Low impact
Ridge Tillage	D	0.12–0.17	Low impact
(Unavailable)	E	0.17–0.22	Low impact
Mulch Tillage	F	0.22–0.26	Medium impact
(Unavailable)	G	0.26–0.31	Medium impact
Reduced Tillage	H	0.31–0.38	Medium impact
(Unavailable)	I	0.38–0.41	Medium impact
(Unavailable)	J	0.41–0.60	Medium impact
Intensive Tillage	K	0.60–1.0	High impact
Mow	MOW	N/A	No impact
Crimp	CRMP	N/A	No impact
Broad-spectrum herbicide	HERB	N/A	No impact

Implements and methods were assigned to a corresponding CULT event within its respective impact category based on STIR score. When multiple CULT events were available to choose from in a given category, Indigo SMEs informed this choice based on knowledge of the given implement / method. Table 7 shows how CULT events were assigned to each implement and method available to growers in the UI. Note that some implements and methods were added to the UI at a later date and were mapped to previously existing implements / methods to ensure consistency with impact category definitions and additional practice requirements; these are nested under the primary implements / methods they are mapped to in Table 7. Note also that the TillageType parameter name used by the DayCent-CR API may be different from the original implement or method name in the Carbon UI.

Table 7: Mapping of CULT events to tillage implements and methods selected in response to the Grower Survey question "What tillage method / equipment was used?"

Tillage Implement or Method	**Impact Category based on STIR**	**DayCent-CR CULT Event**	**Effective depth of disturbance (cm)**	**DayCent-CR TillageType Parameter Name**
Moldboard • Heavy Offset Disk	High	K	30	Intensive Tillage
Deep Ripper • V Ripper • Inline Ripper • Disk Ripper	Medium	H	14	Reduced Tillage
Ridge Till • Lister • Ridge Cultivator	Medium	H	14	Reduced Tillage
Disk • Tandem Disk Harrow	Medium	F	14	Mulch Tillage
Chisel / Chisel Plow • Sweep Plow • Sweep Plow with Cultipacker	Medium	F	14	Mulch Tillage
(Soil) Finisher • Cultipacker • Field Cultivator • Power Tiller • Bedder • Harrow • Rolling Basket	Low	C	6	Growing Season Cultivation
Strip Till	Low	C	6	Growing Season Cultivation
Row Cultivator • Rotary Hoe	Low	C	6	Growing Season Cultivation
Vertical Till	Low	C	6	Growing Season Cultivation

Derivation of the STIR Score

The basis for the additional tillage practice hierarchy, per IndigoCarbon-1_2023_0024a, is the NRCS Soil Tillage Intensity Rating (STIR) score. The STIR score is calculated for each tillage implement or method as follows:

$$STIR = (0.5\ x\ Speed)\ x\ (3.25\ x\ Type)\ x\ Depth\ x\ Area$$

STIR Type relates to the type of mechanical disturbance on the soil that affects the distribution of residue within the soil. Values of STIR Type are given by the NRCS (Lightle, *date unknown*) as:

Table 8: STIR Type used in calculation of STIR score, as given by the NRCS.

STIR Type	**Type of Mechanical Disturbance**
1.0	Inversion, some mixing
0.8	Mixing + some inversion
0.4	Lifting and fracturing
0.7	Mixing only
0.15	Compression

STIR Depth represents the depth to which the residue is incorporated, and does not account for the area being tilled.

Common STIR depths were obtained from the University of Minnesota Extension Upper Midwest Tillage Guide (DeJong-Hughes and Daigh 2021) as well as the NRCS (*Tillage Equipment Pocket Identification Guide* 2017).

STIR Speed represents the typical operation speed of the equipment, as reported by the NRCS (Lightle, *date unknown*), and was obtained for each implement / method from Iowa State University Extension (Hanna 2016) and the University of Arizona (Coates 2002).

STIR Area Disturbed was estimated by Indigo SMEs based on their knowledge of how much of the ground surface is affected by a given implement or method of tillage.

Table 9 contains the required variables needed to calculate the STIR score for each implement or method in Indigo Carbon. Note that some implements and methods were added to the UI at a later date and were mapped to previously existing implements / methods to ensure consistency with impact category definitions and additional practice requirements. The original implements and methods are listed in the table below.

Table 9: STIR score component variables used to calculate the STIR score for the original set of implements and methods in the UI.

Tillage implement/ method	**Speed (mph)**	**STIR Area Disturbed**	**STIR Depth (inch)**	**STIR Type**	**STIR Score**	**Impact Category**
Moldboard	5	1.0	12	1	97.5	High impact
Disk	5	0.70	8	0.8	36.4	Medium impact
Chisel	5	0.70	8	0.8	36.4	Medium impact
Ridge till	6	50%	8	0.7	27.3	Medium impact
Deep Ripper	5	50%	16	0.4	26.0	Medium impact
Vertical till	8	100%	4	0.4	20.8	Low impact
Row Cultivator	7	50%	4	0.7	15.9	Low impact
Finisher	7	50%	4	0.7	15.9	Low impact
Strip-till	8	30%	4	0.7	10.9	Low impact
No-till	0	0%	0	0	0.0	No impact

Derivation of the effective depth of disturbance

An effective depth of disturbance was obtained for each tillage impact category (Table 7). The effective depth of disturbance is used as the basis for the tillage exclusion criteria that requires tillage depths > 20 cm to be excluded from no-till adoption (IndigoCarbon_US-1_2023_0018). The effective depth of disturbance was estimated for each impact category as the average product of the STIR depth and STIR disturbance (i.e. area) for all implements in that category, i.e. within a similar range of STIR scores. This quantity represents the average depth of disturbance across the area being tilled and is therefore different from the STIR depths listed in Table 9. Note that implements were assigned to impact categories based on their complete STIR score, which is a more comprehensive metric for soil disturbance than depth alone.

Special Rule for Tillage: Tillage at Planting for No-Till

In circumstances where a grower is practicing no-till, this requires an artificial tillage event to be submitted at the time of planting using TillageType = No Till - Planter (CULT B). This is so that the effects of the no-till planter are accounted for, which includes a very small amount of soil disturbance as well as movement of carbon between surface and subsurface pools. This rule applies to both cash and cover crop plantings.

Special Rule for Tillage: No Tillage Event Within 30 Days of Planting

A TillageType = No Till - Planter (CULT B) event is specified on the day of planting when no other tillage events occur within the prior 30 days. Thus, CULT B events will occur under reduced till or conventional tillage scenarios

when other forms of tillage were used in the fall and/or in the early spring (i.e., > 30 days prior to planting).

Special Rule for Tillage: Tillage and Harvest Events on the Same Day

Tillage and harvest events cannot coincide on the same day, otherwise DayCent-CR will simulate the tillage event first causing no carbon to be harvested with the harvest event. This will result in artificially high soil carbon. Thus, the events must be separated by one day to ensure harvest is able to remove the proper amount of biomass, leaving the remainder for residue incorporation. In cases where tillage and harvest events occur on the same day, the tillage event is moved to the next available day after harvest, i.e. as soon as no other conflicting events occur.

Special Rule for Tillage: Herbicide to Kill a Crop

Any harvest or termination event that kills a plant requires a harvest event on the day of harvest. With the exception of grain harvests, where "Grain = Yes", this must be followed by a separate tillage event on the day after harvest using TillageType = Broad-spectrum herbicide. As mentioned previously, this type of tillage event does not disturb the soil but is required to move all standing live biomass to the standing dead pool.

Special Rule for Tillage: Root Crops

Any time a root crop is being harvested, a separate tillage event is needed the day after harvest using TillageType = Reduced Tillage (CULT H). This ensures an appropriate amount of soil disturbance for these crops that are harvested using digging equipment. This is needed because the harvest event parameters (Grain and StrawStoverHayRemoval) do not consider digging.

Special Rule for Tillage: Multiple Tillage Events on Same Day

DayCent-CR will not simulate multiple tillage events on the same day, it will only simulate the last event on a given day. To accommodate this limitation, multiple tillage events occurring on the same day must be combined into a single representative event using a mixing efficiency and depth parameter, as described in the DayCent manual (Hartman et al. 2020) on pages 152-3. The total soil disturbance of the events is estimated by computing a combined impact using soil mixing efficiency and depth of tillage, both of which were derived under the guidance of Soil Metrics, LLC (Table 10).

Table 10: Mixing efficiency and multi-event depth used for each implement / method in the calculation of a combined cultivation factor when combining multiple tillage events occurring on the same day.

Tillage Implement or Method	**Mixing Efficiency**	**Multi-Event Depth (mm)**
Moldboard • Heavy Offset Disk	0.95	150
Deep Ripper • V Ripper • Inline Ripper • Disk Ripper	0.35	200
Ridge Till • Lister • Ridge Cultivator	0.47	150
Disk • Tandem Disk Harrow	0.35	150
Chisel / Chisel Plow • Sweep Plow • Sweep Plow with Cultipacker	0.32	150
(Soil) Finisher • Cultipacker • Field Cultivator • Power Tiller • Bedder • Harrow • Rolling Basket	0.24	75
Strip Till	0.18	100
Row Cultivator • Rotary Hoe	0.72	25
Vertical Till	0.18	100

The following assumptions are made in the calculation of a combined cultivation factor, and are taken directly from the DayCent manual (Hartman et al. 2020):

- The impact of multiple tillage events is additive, so the combined tillage effect accounts for previous tillage operations.
- No more than 100% of the soil may be mixed by a tillage operation.
- Once a portion of the soil has been mixed, it cannot be further mixed by additional tillage operations. That is, if 75% of the soil is mixed to 100 mm by one tillage event, a second operation can mix, at most, 25% of the soil in that 100 mm.

A final computed cultivation factor is mapped to the API options available. If a calculated cultivation factor corresponds to an option that is unavailable, the next closest CULT event is chosen within the same impact category as the calculated cultivation factor.

Example Calculation 1. Two field cultivation events (CULT C) on the same day, having a mixing efficiency of 30% and multi-event tillage depth of 50 mm. The combined cultivation factor, CF is:

CF = (0.3 x 50 mm + 0.3 x (50 mm – 0.3 * 50 mm)) / 200 mm = 0.128

The first operation mixes 7.5% of the total soil profile and the second operation about 5.3%, totaling 12.8%. The division by 200 mm comes from the default structure of DayCent and Century, which modeled soil C to only 200 mm. From Table 5.1, 0.128 would be assigned to CULT D.

Example Calculation 2. The second event mixes more intensively and deeper than the first. The first mixes 30% of the soil to a depth of 50 mm, and the second mixes 60% of the soil to a depth of 100 mm. The calculation resembles that in Example Calculation 1, however an additional component to the calculation is included to account for all depths of the soil impacted, not just the depths of overlap as in the first example:

CF = ((0.3 x 50 mm) + (0.6 x 50 mm – (0.3 x 50 mm)) + 0.6 x (100 mm – 50 mm)) / 200mm = 0.3

From Table 5.1, a cultivation factor of 0.3 would be CULT G, however CULT G is not available in the DayCent-CR API. Thus, the next nearest CULT event within the same impact category would be chosen, i.e. CULT H.

Additional events on the same day are added on to the calculation in the same fashion.

Irrigation

Parameter Derivations

The input parameters for irrigation events include IrrigationStartDate, IrrigationEndDate, Auto, and SoilThreshold. IrrigationStartDate and IrrigationEndDate correspond to when the field began irrigation in a given year and when irrigation stopped, respectively. These dates are obtained directly from the Grower Survey questions "When did you start irrigating the field for the season?" and "When did you stop irrigating the field for the season?", respectively.

Because event-level information about irrigation is difficult to obtain, an auto-irrigation routine is used in DayCent-CR that, when turned on, monitors soil water content and triggers irrigation when soil water falls below a threshold. The Auto parameter is set to "Y" (Yes) to indicate that the auto-irrigation routine will be used to control irrigation.

The auto-irrigation routine is turned on to begin monitoring soil water content on the IrrigationStartDate, and it is turned off on the IrrigationEndDate. Note that even though the routine is active for a continuous period of time between the IrrigationStartDate and IrrigationEndDate, this does not mean the field is receiving irrigation water every day. It receives irrigation water only when soil water content falls below the indicated threshold, i.e. the SoilThreshold. The SoilThreshold parameter corresponds to the depletion of available water content below which irrigation is triggered (a fraction), and corresponds to a value below which water stress is expected to occur. In all cases SoilThreshold is set to 0.55. This value was derived from the FAO (Allen et al. 1998) and corresponds well to the row crops found in our program. In summary, when available water content is depleted beyond 55%, irrigation water is applied until field capacity is reached.

Burning

Parameter Derivations

The one input parameter required for burn events is BurnTime, which has three possible options: "No burning", "Yes, before planting", and "Yes, after harvesting". The parameter is set based on the response to the Grower Survey question "Was the crop residue on this field burned?" The question is asked as part of the harvest event in the Carbon UI, and thus corresponds to a BurnTime value of "Yes, after harvesting" when the grower indicates "Yes" to the question. Otherwise the value is set to "No burning". The value "Yes, before planting" is never selected.

Model Spin-up

To support the model spin-up required by DayCent-CR (`IndigoCarbon_US-1_2023_0073`), four historical model input parameters were required. Because this information was difficult to obtain from growers and other sources, a conservative constant value was used instead, based on model run evidence.

Table 11: Spin-up parameters, definitions, available options, and most conservative option based on model run evidence (described below).

Spin-up Parameter	Definition	Options	Conservative Default Value Used in Crediting Runs
Pre-1980	Land use category prior to 1980	Look-up options by LRR, e.g. "Irrigated", "Lowland Non-Irrigated", etc. (Table 8.5)	Upland Non-irrigated
CRP	Land was ever enrolled in the USDA Conservation Reserve Program (Y/N)	Yes, No (Table 8.2)	No
Year1980-2000	Land use category from 1980-2000	Look-up options by LRR, e.g. "Irrigated: Annual Crops in Rotations", etc. (see Table 8.3)	Non-Irrigated: Annual Crops in Rotation
Year1980-2000-Tillage	TillageType input parameter value used from 1980-2000 (IndigoCarbon_US-1_2023_0030)	"Intensive Tillage", "Reduced Tillage", "Mulch Tillage", "Ridge Tillage", "No Tillage", "Growing Season Cultivation", "Mow", "Crimp", "Broad-spectrum herbicide" (Table 8.4)	Intensive Tillage

Model Runs to Establish Conservative Defaults

Model runs were conducted to determine the most conservative option for each spin-up parameter. All options for each parameter were simulated in factorial fashion, varying the geographic location and practice change for a total of ~8,300 practice-changing simulations. Each simulation was 10 years in length, and once completed was compared against a universal baseline simulation to calculate an effect size, analogous to comparing project and baseline simulations in crediting. The baseline was comprised of a rainfed corn-soy-wheat rotation tilled twice per year (spring and fall) and fertilized once (at planting). Effect sizes were then averaged across all simulations within a geographic location. Seventeen geographic locations were used in total, spanning all thirteen of the project LRRs. A corn-soy-spring wheat rotation was used in all simulations. Seven practice changes were tested, including:

- Go from two tillage passes in a year to one
  - Two tillage passes in Spring and Fall of TillageType = Reduced to one in Spring.
- Go from two tillage passes in a year to none
- Decrease tillage impact category (Medium to Low)
- Add a legume cover crop
- Add a non-legume cover crop
- Reduce fertilizer amount by 10%
- Replace 50% of anhydrous ammonia with organic amendment

The most conservative value for each spin-up parameter was determined based on both the minimum effect size as well as the plausibility of a given management option. For the Conservation Reserve Program (CRP) parameter, effect size was minimized in all locations when set to "No", thus "No" was chosen as the default for this parameter.

Table 12: Average effect size estimates (metric tons CO_2e ac^{-1} y^{-1}) for input parameter "CRP" when set to each default option. Results are averaged over all simulations within the given land resource region (LRR) (*Land Resource Regions (LRR)*, 2006). In this exercise some LRRs are split into north (N), south (S), east (E), and west (W) sub-regions, within which results are averaged the same as for LRRs.

LRR	**No**	**Yes**	**Most Conservative**
F	0.0966	0.1015	No
G	0.0961	0.1028	No
HN	0.0956	0.0968	No
HS	0.1002	0.1023	No
IN	0.1587	0.1657	No
IS	0.1285	0.1284	No
J	0.2243	0.2247	No
K	0.1001	0.1069	No
L	0.1489	0.1556	No
M	0.1289	0.1369	No
N	0.1486	0.1509	No
O	0.1495	0.1513	No
PE	0.1523	0.1545	No
PW	0.1767	0.1840	No
R	0.1818	0.1991	No
TE	0.1932	0.1970	No
TW	0.2680	0.2687	No

For the Year1980-2000 parameter, "Non-Irrigated: Annual Crops in Rotation" had the minimum effect size in 13 of 17 locations. Thus "Non-Irrigated: Annual Crops in Rotation" was chosen as the default for this parameter.

Table 13: Average effect size estimates (tons CO_2e ac^{-1} y^{-1}) for input parameter "Year1980-2000" when set to each default option. Empty cells indicate the option was not available for the given location.

LRR	**Irrigated: Annual Crops in Rotation**	**Irrigated: Annual Crops with Hay/Pasture in Rotation**	**Irrigated: Continuous Hay**	**Non-Irrigated: Annual Crops in Rotation**	**Non-Irrigated: Annual Crops with Hay/Pasture in Rotation**	**Non-Irrigated: Continuous Hay**	**Non-Irrigated: Fallow-Grain**	**Most Conservative**
F	-	0.1085	0.1018	0.0966	-	0.1001	0.0951	Non-Irrigated: Fallow-Grain
G	0.1085	0.1065	0.1041	0.0961	-	0.1039	0.0935	Non-Irrigated: Fallow-Grain
HN	0.0987	0.0977	0.0963	0.0956	-	0.0959	0.0956	Non-Irrigated: Fallow-Grain Or Non-Irrigated: Annual Crops in Rotation
HS	0.1058	-	0.1062	0.1002	-	0.1043	0.1015	Non-Irrigated: Annual Crops in Rotation

Table 13: Average effect size estimates (tons CO_2e ac^{-1} y^{-1}) for input parameter "Year1980-2000" when set to each default option. Empty cells indicate the option was not available for the given location.

LRR	Irrigated: Annual Crops in Rotation	Irrigated: Annual Crops with Hay/Pasture in Rotation	Irrigated: Continuous Hay	Non-Irrigated: Annual Crops in Rotation	Non-Irrigated: Annual Crops with Hay/Pasture in Rotation	Non-Irrigated: Continuous Hay	Non-Irrigated: Fallow-Grain	Most Conservative
IN	0.1587	-	0.1917	0.1587	-	0.1840	-	Irrigated: Annual Crops in Rotation Or Non-Irrigated: Annual Crops in Rotation
IS	0.1285	-	0.1345	0.1285	-	0.1304	-	Irrigated: Annual Crops in Rotation Or Non-Irrigated: Annual Crops in Rotation
J	-	-	0.2306	0.2243	-	0.2279	-	Non-Irrigated: Annual Crops in Rotation
K	-	-	0.1103	0.1001	0.1023	0.1109	-	Non-Irrigated: Annual Crops in Rotation
L	-	-	0.1689	0.1489	0.1614	0.1694	-	Non-Irrigated: Annual Crops in Rotation
M	-	-	0.1301	0.1289		0.1303	-	Non-Irrigated: Annual Crops in Rotation
N	-	-	0.1571	0.1486	0.1521	0.1572	-	Non-Irrigated: Annual Crops in Rotation
O	0.1515	-	0.1583	0.1495	-	0.1585	-	Non-Irrigated: Annual Crops in Rotation
PE	-	-	0.1632	0.1523	-	0.1634	-	Non-Irrigated: Annual Crops in Rotation

Table 13: Average effect size estimates (tons CO_2e ac^{-1} y^{-1}) for input parameter "Year1980-2000" when set to each default option. Empty cells indicate the option was not available for the given location.

LRR	Irrigated: Annual Crops in Rotation	Irrigated: Annual Crops with Hay/Pasture in Rotation	Irrigated: Continuous Hay	Non-Irrigated: Annual Crops in Rotation	Non-Irrigated: Annual Crops with Hay/Pasture in Rotation	Non-Irrigated: Continuous Hay	Non-Irrigated: Fallow-Grain	Most Conservative
PW	-	-	0.1997	0.1767	0.1731	0.2002	-	Non-Irrigated: Annual Crops with Hay/Pasture in Rotation
R	-	-	0.2706	0.1818	0.2084	0.2711	-	Non-Irrigated: Annual Crops in Rotation
TE	0.1931	-	0.2056	0.1932	-	0.2062	-	Irrigated: Annual Crops in Rotation
TW	0.2680	-	0.2728	0.2680	0.2673	0.2733	-	Non-Irrigated: Annual Crops with Hay/Pasture in Rotation

For the Year1980-2000-Tillage parameter, "Intensive Tillage" had the minimum effect size in 14 of 17 locations. Thus "Intensive Tillage" was chosen as the default for this parameter.

Table 14: Average effect size estimates (tons CO_2e ac^{-1} y^{-1}) for input parameter "Year1980-2000 Tillage" when set to each default option. Empty cells indicate the option was not available for the given location

LRR	Intensive Tillage	No Till	Reduced Tillage	Most Conservative
F	0.0962	0.1005	0.0966	Intensive Tillage
G	0.0961	0.1045	0.0961	Intensive Tillage
HN	0.0955	0.0960	0.0956	Intensive Tillage
HS	0.0994	0.1032	0.1002	Intensive Tillage
IN	0.1576	0.1648	0.1587	Intensive Tillage
IS	0.1288	0.1285	0.1285	No Till
J	0.2248	0.2240	0.2243	No Till
K	0.0928	0.1132	0.1001	Intensive Tillage
L	0.1489	0.1708	0.1489	Intensive Tillage
M	0.1242	0.1407	0.1289	Intensive Tillage
N	0.1482	0.1521	0.1486	Intensive Tillage
O	0.1485	0.1528	0.1495	Intensive Tillage
PE	0.1513	0.1562	0.1523	Intensive Tillage
PW	0.1707	0.1863	0.1767	Intensive Tillage
R	0.1676	0.2207	0.1818	Intensive Tillage
TE	0.1901	0.1987	0.1932	Intensive Tillage
TW	0.2684	0.2681	0.2680	Reduced Tillage

For the Pre-1980 parameter, "Upland Non-Irrigated" had the lowest effect size in all 17 locations, in many cases tying

"Lowland Non-Irrigated", a less-plausible historical management option given that the cropping systems currently in the project primarily occur on upland soils. Thus "Upland Non-Irrigated" was chosen as the default for this parameter. Note that "Livestock Grazing" was omitted from the list of plausible options for Pre-1980 on the basis of its rarity on cropland acres after the Green Revolution (Smart et al. 2019). Grazing is also not included in the project at this time, and is not validated for modeling.

Table 15: Average effect size estimates (tons CO_2e ac^{-1} y^{-1}) for input parameter "Pre-1980" when set to each default option. Empty cells indicate the option was not available for the given location

LRR	Irrigation (Pre 1980s)	Lowland Non-Irrigated (Pre 1980s)	Upland Non-Irrigated (Pre 1980s)	Most Conservative
F	0.0991	0.0966	0.0966	Upland Non-Irrigated
G	0.1090	-	0.0961	Upland Non-Irrigated
HN	0.0959	0.0956	0.0956	Upland Non-Irrigated
HS	0.1015	0.1002	0.1002	Upland Non-Irrigated
IN	0.1597	0.1587	0.1587	Upland Non-Irrigated
IS	0.1287	0.1285	0.1285	Upland Non-Irrigated
J	-	0.2243	0.2243	Upland Non-Irrigated
K	-	0.1001	0.1001	Upland Non-Irrigated
L	-	0.1489	0.1489	Upland Non-Irrigated
M	-	0.1289	0.1289	Upland Non-Irrigated
N	-	0.1486	0.1486	Upland Non-Irrigated
O	0.1497	0.1495	0.1495	Upland Non-Irrigated
PE	-	0.1523	0.1523	Upland Non-Irrigated
PW	-	0.1767	0.1767	Upland Non-Irrigated
R	-	0.1818	0.1818	Upland Non-Irrigated
TE	0.1941	0.1932	0.1932	Upland Non-Irrigated
TW	0.2695	0.2680	0.2680	Upland Non-Irrigated

Non-Management-Based Inputs

Weather Inputs

The DayCent-CR API requires meteorological inputs be provided in a weather file containing the following tab-delimited columns of daily weather data:

Column 1 - Day of month, 1-31

Column 2 - Month of year, 1-12

Column 3 - Year (4 digits)

Column 4 - Day of the year, 1-366

Column 5 - Maximum temperature for day, ^oC

Column 6 - Minimum temperature for day, ^oC

Column 7 - Precipitation for day, cm

Column 8 - Solar radiation, langleys day^{-1}

Column 9 - Relative humidity, %, 1-100

Column 10 - Wind speed, miles per hour

Each field in the project has a corresponding weather file constructed for use in any simulation for that field. Each weather file begins on January 1, 1975, and runs through the end of the final project period year. Leap years are accounted for such that February 29 in leap years is present. The file format is tab-delimited with no header, and compressed using XZ compression. The file extension is ".wth.xz". All files are uploaded to the Google Cloud bucket provided by Soil Metrics.

While only columns 1-7 are required by the API, and columns 8-10 are optional, all ten columns are provided for crediting simulations. The source of temperature, precipitation, solar radiation, relative humidity, and wind speed data is the North American Land Data Assimilation System (NLDAS), a gridded data product produced by the National Aeronautics and Space Administration (NASA). (See IndigoCarbon_US-1-2020-0069 for a description of NLDAS). Hourly data from NLDAS are aggregated to a localized day according to a field's time zone. The earliest available NLDAS hour is January 1, 1979 13:00, thus the earliest complete day available is January 2, 1979 since

NLDAS is provided in Coordinated Universal Time (UTC). Thus January 2, 1979 is copied to fill January 1, 1979. Similarly, January 1, 1979- December 31, 1982 is copied to fill in January 1 1975-Dec 31, 1978, since these data are required but not available with NLDAS.

Soil Inputs (Data Conversions)

The following soil input parameters are required by DayCent-CR at each modeled location:

- Soil sample "date"
- Soil sample "name"
- Soil sample "hydric" (Y/N)
- Soil sample "initialsocstock30cm"

The following soil input parameters are required by DayCent-CR at each modeled location for each soil layer (Table 16):

- "BulkDens"
- "CoarseFragments"
- "Sand"
- "Clay"
- "OrgFrac"
- "Ph"

Table 16: Layer depths for soil input parameters.

Layer #	**Top Depth of Layer (cm)**	**Bottom Depth of Layer (cm)**
1	0	2
2	2	5
3	5	10
4	10	20
5	20	30
6	30	45
7	45	60
8	60	75
9	75	90
10	90	105
11	105	120
12	120	150
13	150	180
14	180	210

The soil data information required by DayCent-CR is listed in Tables 17 and 18. Some inputs such as "date" and "name" are only required once per location, but other inputs such as bulk density and sand fraction are required for each modeled layer. Inputs that are layer-specific are typically set differently for the first 30 cm of soil than for the next 180 cm of soil. Most will be set uniformly for the first 30 cm using direct measurements or information derived from measurements, such as imputed bulk density used to calculate SOC stock (IndigoCarbon_US-1-2020-0054). For layers below 30 cm, inputs are typically set using the corresponding available gSSURGO properties for that location and soil layer. (See IndigoCarbon_US-1-2020-0069 for a description of gSSURGO.) In the event that a gSSURGO layer overlaps a required DayCent-CR layer, the weighted average of the gSSURGO layers that overlap the DayCent-CR layer is used.

Table 17: Soil input parameters that do not vary with soil layer.

Parameter Name	**Definition**	**Unit**	**Data Conversions**
date	Date of the soil sample measurement	MM/DD/YYYY	No conversion needed, however if multiple samples are taken, the earliest date is used in order to support the model spin-up (IndigoCarbon_US-1_2023_0073)
name	Unique identifier for a soil sample location	<string>	N/A
hydric	Boolean for the presence of hydric soil, as defined by the USDA NRCS (*Hydric Soils - Introduction*, 2021)	<Boolean>	This value is True if either the hydricating component is labeled in gSSURGO, or the weighted average of gSSURGO organic matter (OM) for layers down to 30 cm is > 10%. The hydric and OM values from gSSURGO are determined using the major component of each property within the map unit that contains the soil sample location. In cases where gSSURGO data are missing, False is assumed.
Initialsocstock30cm	SOC stock for top 30 cm of soil	g m^{-2} in top 30 cm	Calculated as the product of soil carbon concentration (SOC, %) and bulk density (BD, g cm^{-3}) obtained from IndigoCarbon_US-1_2023_0054: initialsocstock30cm = BD x SOC x 30 x 10,000 The factors 30 and 10,000 are needed to obtain stock for the top 30 cm and convert from cm^2 to m^2, respectively.

For soil input parameters that vary with soil layer, the logic outlined in Table 17 is used for all soil layers in the top 30 cm. For layers deeper than 30 cm, gSSURGO data are used corresponding to the major gSSURGO component for that sample location and layer depth. If no gSSURGO data are available beneath a specific layer, the deepest available gSSURGO layer is repeated as many times as needed to create a 210 cm soil profile. If no gSSURGO data are available at or above a specific layer, the gSSURGO data are repeated from below up to 30 cm. If no gSSURGO data are available at a specific location in any layer, values are taken from the first 30 cm and repeated down through the profile to 210 cm.

Table 18: Soil input parameters that vary with soil layer.

Parameter Name	Definition	Unit	Data Conversions
BulkDens	Bulk density in top 30 cm	g cm^{-3}	Derived from Equation 7 Saxton et al. (1986) in conjunction with measured sand and clay fractions. Note that this value of bulk density differs from the measured value used to compute soil carbon stock. See *Bulk Density* section below.
CoarseFragments	Percent of soil mass in a given layer that is in coarse fragments	%	The gSSURGO property "fagvol_r" is used for each soil layer, where available. If missing, 0 is assumed. All fractions of rock fragments > 2 mm are summed.
Sand	Percent of layer that is sand	%	<None>
Clay	Percent of layer that is clay	%	<None>
OrgFrac	Percent of layer that is organic matter	%	For layers above 30 cm, this is calculated as SOC concentration divided by 0.58, the von Bemmelen approximation (Heaton et al. 2016) For layers below 30 cm, the gSSURGO value of organic matter percentage.
Ph	pH	<unitless>	<None>

Bulk Density

An estimate of bulk density is required in the derivation of two independent model input parameters:

1. "***Initialsocstock30cm***" (soil carbon stock), calculated as the product of measured SOC concentration, depth, and measured bulk density, per the SEP Section 6.5;
2. "***BulkDens"***, a value of bulk density used in DayCent-CR's soil hydrologic processes. This parameter is not used in the calculation of initial soil carbon stock (Initialsocstock30cm) nor is it used in dynamic calculations of soil carbon stock in DayCent-CR 1.0.

As mandated by the SEP, the soil carbon measurement is calculated for ***Initialsocstock30cm*** using only measured values of SOC concentration and bulk density. The use of measured bulk density ensures soil carbon mass balance is calculated correctly for each modeled location.

The ***BulkDens*** parameter is calculated using measurements of soil texture in Equation 7 in Saxton et al. (1986), as described below. The ***BulkDens*** parameter is independent from ***Initialsocstock***, and is only used to support the soil hydrology computations in DayCent-CR; it has no direct impact on simulations of soil carbon. Measured bulk density was not used for this parameter because DayCent-CR has not been developed to use direct bulk density measurements in soil hydrologic processes. Rather, the traditional procedure used by Soil Metrics, the model's developers, has been to obtain ***BulkDens*** using an equation (Equation 7) from Saxton et al. (1986). This is also the approach used in calibrating and validating DayCent-CR 1.0 (IndigoCarbon_US-1_2023_0046). The use of measured bulk density from the Project was tested in collaboration with Soil Metrics, but revealed that measured bulk density was often outside of the tolerable limits for the ***BulkDens*** parameter, causing simulations to fail unless ***BulkDens*** was tightly bounded. Thus there is a need for further model development before measured bulk density may be used to support soil hydrology in DayCent-CR. Until then, ***BulkDens*** should be derived from Equation 7 of Saxton et al. (1986) to ensure optimal performance of the validated model. The BulkDens parameter is derived as follows, based on Equation 7 in Saxton et al. (1986) for saturated soil moisture content (sat). Sat is calculated as a function of sand and clay content, using the ***Sand*** and ***Clay*** input parameters listed in Table 18

$$sat = \ 0.332 - 7.251e^{-4} \ \times \ Sand + 0.1276 \ (\log_{10} Clay)$$

BulkDens is then calculated by multiplying (1 – *sat*) by the dry particle density of soil, 2.65 g cm^{-3}.

$$BulkDens = 2.65 \ (1 - sat)$$

References

Allen, R. G., Pereira, L. S., Raes, D., & Smith, M. (1998). *Crop Evapotranspiration: Guidelines for Computing Crop Water Requirements*. Food and Agriculture Organization of the United Nations. https://www.fao.org/3/X0490E/x0490e00.htm#Cor

Bolinder, M. A., et al. "Estimating C inputs retained as soil organic matter from corn (Zea mays L.)." *Plant and Soil* 215.1 (1999): 85-91.

Coates, W. E. (2002). *Agricultural Machinery Management*. University of Arizona. https://cals.arizona.edu/crops/equipment/agma

DeJong-Hughes, J., & Daigh, A. (2021). *Upper Midwest Tillage Guide*. University of Minnesota Extension. https://drive.google.com/file/d/0B3cuyvdDn64PRUd4TmpfVllKLXJia29JMDFVcE1UQ0dGLUhr/view?resourcekey=0-8UdTGAONCp3iEe4lH8YZaQ

Hanna, M. (2016). *Estimating the Field Capacity of Farm Machines*. Iowa State University Extension and Outreach. https://store.extension.iastate.edu/Product/Estimating-Field-Capacity-of-Farm-Machines-pdf

Hartman, M., Parton, W. J., del Grosso, S. J., Easter, M., Hendryx, J., Hilinski, T., Kelly, R., Keough, C. A., Killian, K., Lutz, S., Marx, E., McKeown, R., Ogle, S., Ojima, D. S., Paustian, K., Swan, A., & Williams, S. (2020). *The Daily Century Ecosystem, Soil Organic Matter, Nutrient Cycling, Nitrogen Trace Gas, and Methane Model: User Manual, Scientific Basis, and Technical Documentation*. Colorado State University.

Heaton, Lee, Michael A. Fullen, and Ranjan Bhattacharyya. "Critical analysis of the van Bemmelen conversion factor used to convert soil organic matter data to soil organic carbon data: comparative analyses in a UK loamy sand soil." *Espaço Aberto* 6.1 (2016): 35-44.

Hydric Soils - Introduction. (2021). USDA Natural Resources Conservation Service. Retrieved 2021, from https://www.nrcs.usda.gov/wps/portal/nrcs/detail/soils/use/hydric/?cid=nrcs142p2_053961

Karlen, Douglas L. and Huggins, David R., "Crop Residues" (2014). Publications from USDA-ARS / UNL Faculty. 1436. https://digitalcommons.unl.edu/usdaarsfacpub/1436

K. E. Saxton, et al. "Estimating Generalized Soil-water Characteristics From Texture." *Soil Science Society of America journal*, v. 50 ,.4 pp. 1031-1036.

Lafond, G. P., et al. "Quantifying straw removal through baling and measuring the long-term impact on soil quality and wheat production." *Agronomy Journal* 101.3 (2009): 529-537.

Land Resource Regions (LRR). (2006). USDA Natural Resources Conversation Service. https://www.nrcs.usda.gov/wps/portal/nrcs

Lightle, D. T. *Soil Tillage Intensity Rating STIR*. USDA NRCS. https://www.mssoy.org/uploads/files/stir-narrative-three.pdf

Mayer, M. (2012). Placing a Value on Corn Stover. UW Extension. https://green.extension.wisc.edu/files/2010/05/Placing-a-Value-on-Corn-Stover.pdf

Roth, G. W. (2016). *Considerations in Managing Cutting Height of Corn Silage*. PennState Extension. https://extension.psu.edu/considerations-in-managing-cutting-height-of-corn-silage

Saxton, K. E., Rawls, W. J., Romberger, J. S., Papendick, R. I. 1986. "Estimating Generalized Soil-water Characteristics From Texture." *Soil Science Society of America journal*, v. 50 ,.4 pp. 1031-1036.

Smart, A., D. Redfearn, R. Mitchell, T. Wang, C. Zilverberg, P. J. Bauman, J. D. Derner, J. Walker and C. Wright. 2019. Forum: Integration of Crop-Livestock Systems: An Opportunity to Protect Grasslands from Conversion to Cropland in the US Great Plains. Rangeland Ecology and Management.

Sustainable Corn Stover Harvest. (2013). Iowa Corn Promotion Board. https://www.iowacorn.org/media/cms/IowaCornResearchBrochure_Final_IFT_F4B608A12ED16.pdf

Swan, A., Easter, M., Chambers, A., Brown, K., Williams, S. A., Creque, J., Wick, J., & Paustian, K. (2020, July). *COMET-Planner: Carbon and Greenhouse Gas Evaluation for NRCS Conservation Practice Planning*. USDA-NRCS, Colorado State University.

Tillage Equipment Pocket Identification Guide. (2017). Natural Resources Conservation Service. https://www.nrcs.usda.gov/Internet/FSE_DOCUMENTS/nrcseprd1323703.pdf

2014 Farm Bill - Conservation Compliance Crop List | NRCS. (2014). USDA Natural Resources Conservation Service. https://www.nrcs.usda.gov/wps/portal/nrcs/detail/national/programs/farmbill/?cid=stelprdb1262733

U.S. Environmental Protection Agency. 2021. Inventory of U.S. Greenhouse Gas Emissions and Sinks. 1990-2019. https://www.epa.gov/sites/default/files/2021-04/documents/us-ghg-inventory-2021-main-text.pdf?

Xu, Hui, et al. "A global meta-analysis of soil organic carbon response to corn stover removal." *Gcb Bioenergy* 11.10 (2019): 1215-1233.

2 Summary of Changes

Section	Date	Summary of Changes
Tillage Parameter Derivations	June 25, 2024	CULT B is used at planting to match DayCent-CR Version 1.1.0 calibration approach
Special Rule for Tillage: No Tillage Event Within 30 Days of Planting	June 25, 2024	Added new rule to match DayCent-CR Version 1.1.0 calibration approach